

Sprint 1 Demo -- IBMS (Integrated Business Management Suite)

Field	Detail
Project Name	Integrated Business Management Suite (IBMS) v2.0
Team	Shivansh Srivastava (Product Developer), Prahallad Padhan (Product Owner), Ranveer Rai Khare (Scrum Master)
Sprint	Sprint 1
Demo Date	April 12, 2026
Duration	~15 minutes
Environment	Local (localhost:8000) -- FastAPI + MongoDB 8.2 + Vanilla JS SPA

1. Demo Objective

Demonstrate all 10 user stories delivered in Sprint 1 by walking through the running IBMS application, exercising the API endpoints, and showing the live frontend dashboard.

Sprint Goal Recap: Build a production-grade, AI-first enterprise management platform with real-time dashboards, AI-powered analytics, enterprise-grade security, and multi-environment deployment support.

2. Pre-Demo Checklist

Step	Command / Action	Expected State
1	Start MongoDB: <code>`net start MongoDB`</code>	MongoDB running on port 27017
2	Activate venv: <code>`.venv2\Scripts\activate`</code>	Python 3.12 venv active
3	Start server: <code>`uvicorn server:app --host 0.0.0.0 --port 8000 --reload`</code>	Server running at <code>`http://localhost:8000`</code>
4	Open browser: <code>`http://localhost:8000`</code>	IBMS login screen renders
5	Verify MongoDB: check console for <code>`MongoDB async connected`</code>	Database connected

3. Demo Script -- Feature Walkthrough

Demo 1: User Registration & Secure Login (US-01) -- 2 min

Step	Action	What to Show
1.1	Open `http://localhost:8000` in browser	Auth overlay with Login/Register toggle
1.2	Click "Register" tab	Registration form appears
1.3	Type a weak password (e.g., `abc`)	Password strength meter shows Weak (red)
1.4	Type a strong password (e.g., `Secure@2026!`)	Strength meter shows Strong (green)
1.5	Fill in username, email, password -> Submit	Success toast -> auto-login -> Dashboard
1.6	API Demo: Show `POST /api/auth/register` response in DevTools	JWT access token + CSRF token returned
1.7	Show MongoDB `users` collection	Password stored as bcrypt hash (`\$2b\$12\$...`), not plaintext

Acceptance Criteria Validated: Password strength validation, bcrypt hashing, JWT issuance, device fingerprint capture, frontend overlay.

Demo 2: Two-Factor Authentication (US-02) -- 1.5 min

Step	Action	What to Show
2.1	Call `POST /api/auth/2fa/setup` with Bearer token	Returns TOTP provisioning URI + secret
2.2	Show provisioning URI is compatible with authenticator apps	URI format: `otpauth://totp/IBMS:user@...`
2.3	Call `POST /api/auth/2fa/confirm` with a valid TOTP code	2FA enabled successfully
2.4	Logout and log back in	Login form now shows the 2FA code input field
2.5	Enter TOTP code -> Login	Login succeeds with 2FA
2.6	Call `POST /api/auth/2fa/disable`	2FA disabled, login reverts to password-only

Acceptance Criteria Validated: TOTP setup/confirm/disable, dynamic 2FA field on login, provisioning URI generation.

Demo 3: Role-Based Dashboard Access (US-03) -- 1.5 min

Step	Action	What to Show
3.1	Call `GET /api/auth/roles`	Returns 5 roles with permission arrays
3.2	Login as `admin` user -> call `GET /api/auth/me`	Shows resolved permissions for admin role
3.3	Show sidebar -- all 11 pages visible	Admin sees all navigation items
3.4	Login as `viewer` user	Sidebar shows only Dashboard and basic pages
3.5	As viewer, call `GET /api/audit/log`	Returns `403 Forbidden` (requires `audit.view`)
3.6	As admin, call `GET /api/audit/log`	Returns paginated audit events

Acceptance Criteria Validated: 5-tier RBAC, permission inheritance, sidebar visibility, 403 on unauthorized access.

Demo 4: Real-Time KPI Dashboard (US-04) -- 2 min

Step	Action	What to Show
4.1	Navigate to Dashboard page	9 KPI cards with animated counters load
4.2	Point out the WebSocket status indicator	Green dot = connected to `/ws/kpi`
4.3	Wait 15 seconds	KPI values update automatically (live push)
4.4	Open DevTools -> Network -> WS tab	Show WebSocket frames arriving every 15s
4.5	Call `GET /api/dashboard`	JSON snapshot: revenue, margin, risk, compliance, etc.
4.6	Call `GET /api/dashboard/history`	Historical data points for trend charts
4.7	Navigate to Analytics page	Chart.js KPI history trend chart renders
4.8	Show 3D tilt spotlight effect on KPI cards	Hover over cards -> GPU-accelerated tilt

Acceptance Criteria Validated: Live KPI push (15s), WebSocket auto-reconnect, Chart.js charts, Redis/in-memory cache, animated counters.

Demo 5: AI-Powered Business Insights & Copilot (US-05) -- 1.5 min

Step	Action	What to Show
5.1	Call <code>`GET /api/ai/insights`</code>	4 insights: anomaly, trend, prediction, recommendation
5.2	Call <code>`GET /api/ai/anomalies`</code>	Active anomalies in transaction volume
5.3	Call <code>`POST /api/copilot/ask`</code> with <code>`{"query":"show risk details"}`</code>	AI returns risk metrics from KPI data
5.4	Call <code>`POST /api/forecast`</code>	Prophet Ensemble v2 forecast with confidence bands
5.5	Navigate to AI Insights page in frontend	AI insight cards rendered in the UI

Acceptance Criteria Validated: AI insights (4 types), anomaly detection, NL copilot, forecast with CI bands, recommendations.

Demo 6: Risk Scoring & Fraud Detection (US-06) -- 1.5 min

Step	Action	What to Show
6.1	<code>`POST /api/risk/score`</code> with <code>`{"transaction":{"amount":75000,"vendor":"new"}}`</code>	Risk score 0-100 with factor breakdown
6.2	<code>`POST /api/risk/composite`</code> with <code>`{"factors":{"amount":80,"behavior":50,"compliance":30}}`</code>	Weighted score: $(0.5 \times 80) + (0.3 \times 50) + (0.2 \times 30) = 61.0$
6.3	<code>`POST /api/fraud/detect`</code> with <code>`{"transaction":{"amount":100000,"vendor":"unknown"}}`</code>	Fraud score (0-1) + <code>`flag_for_review: true`</code>
6.4	<code>`POST /api/decision/evaluate`</code> with a high-risk input	Decision: approve / review / reject
6.5	Navigate to Risk & Fraud page in frontend	Risk scoring form + results displayed

Acceptance Criteria Validated: Transaction risk (0-100), composite weighted risk, Isolation Forest fraud, decision routing.

Demo 7: Compliance Checking & Audit Trail (US-07) -- 1 min

Step	Action	What to Show
7.1	`POST /api/compliance/check` with `{"transaction":{"amount":600000}}`	`passed: false`, violation: `MISSING_HIGH_VALUE_APPROVAL`
7.2	`POST /api/compliance/check` with `{"transaction":{"amount":600000,"approval_ref":"APR-001"}}`	`passed: true`, no violations
7.3	`GET /api/audit/log` (as admin)	Paginated audit events with timestamps
7.4	Show audit entries in MongoDB `audit_logs` collection	Events with base64-encoded details

Acceptance Criteria Validated: Control-set validation, violation codes, audit trail persistence, RBAC-gated audit log.

Demo 8: Budget Optimization & Dynamic Pricing (US-08) -- 1 min

Step	Action	What to Show
8.1	`POST /api/budget/optimize` with `{"items":[{"name":"Marketing","amount":50000}, {"name":"R&D","amount":80000}], "growth_target":15}`	Optimized totals with growth multiplier
8.2	`POST /api/pricing/suggest` with `{"demand_index":0.8,"stock_level":200,"competitor_price":999}`	Recommended dynamic price
8.3	Navigate to Budget and Pricing pages in frontend	Forms + result output displayed

Acceptance Criteria Validated: Budget optimizer with growth target, dynamic pricing with demand/stock/competitor weights.

Demo 9: Lead Scoring & Inventory Prediction (US-09) -- 1 min

Step	Action	What to Show
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Step	Action	What to Show
9.1	<code>`POST /api/leads/score` with `{"engagement":{"email_opens":15,"page_visits":30,"demo_requests":2},"fit":{"company_size":"enterprise","industry":"fintech","budget":500000}}`</code>	Score + tier (hot/warm/cold)
9.2	<code>`POST /api/inventory/predict` with `{"product_id":"SKU-001","lead_time_days":14}`</code>	Reorder point prediction
9.3	Navigate to Lead Scoring page in frontend	Lead scoring form + tier output

Acceptance Criteria Validated: Lead qualification (55% engagement + 45% fit), tier classification, inventory reorder prediction.

Demo 10: System Health Monitoring & Deployment (US-10) -- 2 min

Step	Action	What to Show
10.1	<code>`GET /api/health`</code>	<code>`status: healthy`</code> , uptime, request count, error rate
10.2	<code>`GET /api/metrics`</code>	System metrics snapshot
10.3	<code>`GET /api/system/status`</code>	Top endpoints, cache state, detailed status
10.4	Show response headers in DevTools	<code>`X-Trace-Id`</code> , <code>`X-Response-Time-Ms`</code> , security headers
10.5	Navigate to System Health page in frontend	Server status dashboard
10.6	Show <code>`docker-compose.yml`</code>	Nginx + Web + Redis + MongoDB orchestration
10.7	Show <code>`deploy/k8s/ibms-core-deployment.yaml`</code>	3 replicas, resource limits, health probes
10.8	Show <code>`deploy/aws/terraform/main.tf`</code>	VPC, subnets, ElastiCache, security groups

Acceptance Criteria Validated: Health/metrics/status endpoints, trace headers, security headers, Docker/K8s/Terraform configs.

4. Live Database Verification

Collection	Documents	Key Data
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Collection	Documents	Key Data
`users`	3	admin, analyst, shiv1 -- bcrypt-hashed passwords
`kpi_snapshots`	24	Revenue, margin, risk, compliance -- every 15s
`kpi_latest`	1	Materialized view -- latest KPI for instant reads
`audit_logs`	2+	Login, register events -- timestamped, base64
`refresh_tokens`	1+	Active refresh tokens with TTL expiry index
`csrf_tokens`	1+	CSRF tokens with 1-hour TTL expiry index
`notifications`	1+	System notifications
`alerts`	0	Ready -- indexed by severity + resolved status
`ai_recommendations`	0	Ready -- indexed by company + recommendation type
`enterprise_profiles`	0	Ready -- unique index on company_id
`webhook_logs`	0	Ready -- indexed by processed status
`decision_rules`	0	Ready -- indexed by module + enabled status
`rate_limits`	0+	IP-based rate limit tracking

Total Collections: 13 with production-ready indexes and TTL policies.

5. Demo Summary -- Sprint 1 Delivered Features

#	Feature Area	Endpoints Demonstrated	Story Points	Verdict
1	Registration & Login	5 endpoints	8	[x] Fully Working
2	Two-Factor Authentication	3 endpoints	5	[x] Fully Working
3	Role-Based Access Control	3 endpoints	8	[x] Fully Working
4	Real-Time KPI Dashboard	3 endpoints + WS	8	[x] Fully Working
5	AI Insights & Copilot	4 endpoints	8	[x] Fully Working
6	Risk Scoring & Fraud Detection	4 endpoints	8	[x] Fully Working
7	Compliance & Audit Trail	2 endpoints	5	[x] Fully Working
8	Budget & Dynamic Pricing	2 endpoints	5	[x] Fully Working
9	Lead Scoring & Inventory	2 endpoints	5	[x] Fully Working

#	Feature Area	Endpoints Demonstrated	Story Points	Verdict
10	System Health & Deployment	3 endpoints + configs	8	[x] Fully Working
	Total	31+ endpoints	68 pts	10/10 Done

6. Q&A / Stakeholder Feedback

Questions anticipated:

Question	Answer Summary
How is data persisted?	MongoDB 8.2 (13 collections, TTL indexes, materialized views)
What if Redis goes down?	Graceful fallback to in-memory dict -- zero downtime
How are tokens secured?	JWT HS256 (30-min TTL) + refresh rotation (7-day) + device binding
What ML model is used for fraud?	Isolation Forest (scikit-learn) -- unsupervised, no labeled data needed
Can this scale horizontally?	K8s 3-replica deployment ready; stateless backend allows scaling
What's the test coverage?	14 tests (5 unit + 9 integration), all passing -- improving Sprint 2

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